

## Fall 25, 4100 Problem Set 6

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given  $A = \begin{bmatrix} 2 & -4 & 1 & 3 \\ 3 & -6 & 0 & -3 \\ -1 & 2 & 1 & 0 \\ 4 & -8 & -1 & 3 \end{bmatrix}$  compute  $\text{adj}(A)$  and use it to find bases for  $\text{null}(A)$  and  $\text{null}(A^T)$

2. Find the eigenvalues and eigenvectors given the following matrices:

a.  $A = \begin{bmatrix} 4 & 3 \\ 5 & 6 \end{bmatrix}$

b.  $B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 4 \\ 0 & 2 & -2 \end{bmatrix}$

3. Let  $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$  be such that for linear independent  $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{C}^n$  we have that  $T\mathbf{v}_1 = \mathbf{v}_1 + 6\mathbf{v}_2$  and  $T\mathbf{v}_2 = 3\mathbf{v}_1 + 4\mathbf{v}_2$ . Find two eigenvalues and associated eigenvectors of  $T$ .

4. Let  $A$  and  $B$  be similar such that  $A = M^{-1}BM$ . Prove that  $A$  and  $B$  have the same characteristic and minimal polynomial.

5. Find the set of all  $A \in \mathbb{R}^{2 \times 2}$  such that  $A^6 = I$  up to similarity classes by finding all possible minimal polynomials.

6. Let  $A \in \mathbb{R}^{n \times n}$  be invertible and have characteristic polynomial of  $p(\lambda)$ . Show that  $A^{-1}$  will have characteristic polynomial of

$$q(\lambda) = \frac{(-\lambda)^n}{|A|} p\left(\frac{1}{\lambda}\right)$$